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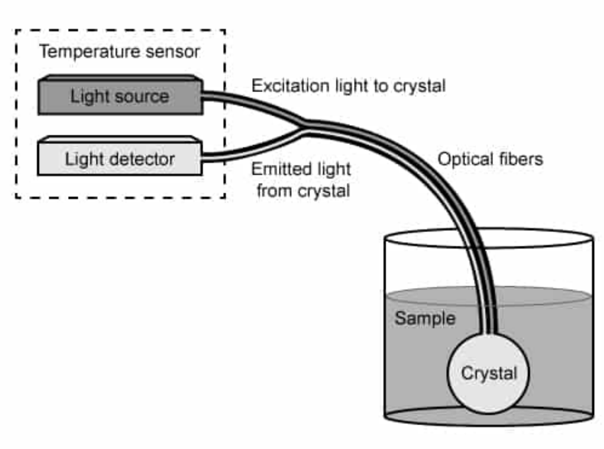

Figure 1 Schematic of a fiber optic temperature probe

Light is transmitted along continuous glass fibers that are approximately the width of a human hair. The fiber consists of a core surrounded by a cladding with a lower refractive index than the core. The difference in refractive indexes causes total internal reflection of the light at the interface between the core and the cladding, minimizing the loss of light intensity even over relatively long distances. One bundle of optical fibers transmits the excitation light to the crystal, and a separate bundle of optical fibers transmits the emitted light back to the device for analysis.

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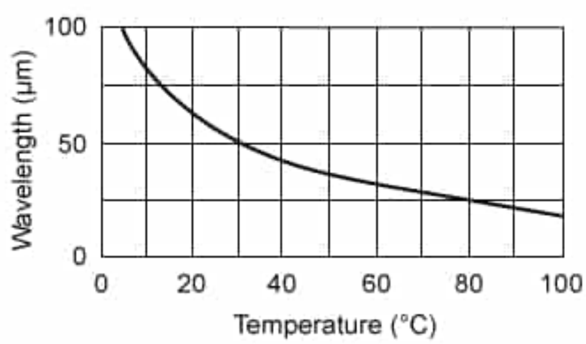


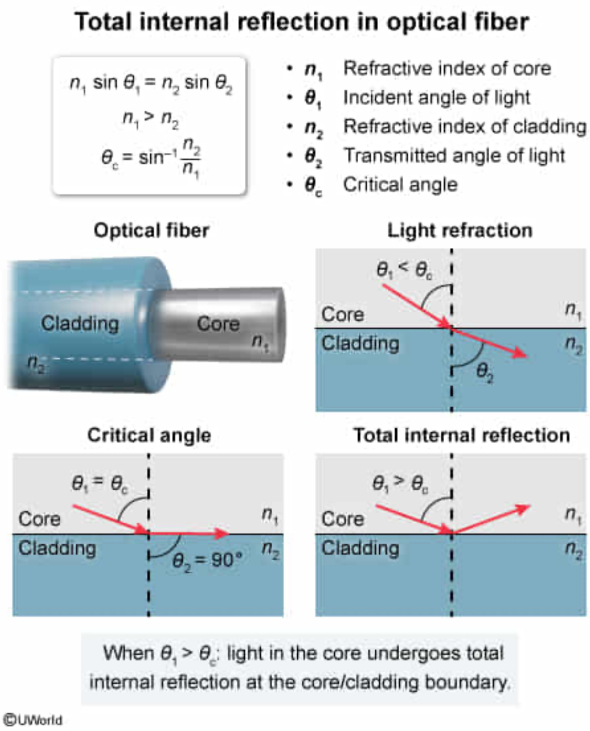
Figure 2 Plot of the infrared wavelength emitted at different temperatures

Which of the following statements best explains why light travels only in the core of an optical fiber?

- A. There is no light refraction at the boundary between the core and cladding. [13%]
- B. The speed of light is greater in the core than in the cladding due to differences in the refractive indexes. [9%]
- C. The incident angle of the light at the boundary between the core and cladding is less than the critical angle. [27%]
- ✔ D. The incident angle of the light at the boundary between the core and cladding is greater than the critical angle. [49%]

Correct
48% answered correctly

Explanation:



Refraction is the bending of a light beam when it travels from a material with a refractive index of n_1 into a material with a refractive index of n_2. The incident angle θ_1 of the light with respect to the normal of the material boundary is related to the transmitted angle θ_2 of the light by **Snell's law**:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

The **critical angle** θ_c is defined as the value of θ_1 that results in θ_2 equaling 90°, which causes the refracted light to travel along the boundary between the two materials. In this situation:

$$\sin\theta_2 = \sin 90° = 1$$

The first equation can be rearranged to solve for θ_c, giving:

$$\theta_c = \sin^{-1}\frac{n_2 \sin\theta_2}{n_1}$$

$$\theta_c = \sin^{-1}\frac{n_2 \sin 90°}{n_1}$$

$$\theta_c = \sin^{-1}\frac{n_2}{n_1}$$

The inverse sine function is only defined for values between −1 and 1. Therefore, n_1 must be greater than n_2 for the above equation to be valid:

$$n_1 > n_2$$

When θ_1 is greater than θ_c, refraction causes all the light to be reflected from the boundary, achieving total internal reflection.

In this question, light travels from the core of the optical fiber into the cladding. The refractive index n_1 of the core is greater than the refractive index n_2 of the cladding:

$$n_1 > n_2$$

Therefore, light travels in only the core of the fiber when the incident angle θ_1 is greater than the critical angle θ_c:

$$\theta_1 > \sin^{-1}\frac{n_2}{n_1}$$

(Choice A) Light refraction does occur at the boundary between the core and the cladding because the two materials have different values of *n*.

(Choice B) The refractive index of the core is greater than the refractive index of the cladding. Therefore, the speed of light is less in the core than in the cladding, not greater.

(Choice C) Total internal reflection occurs when the incident angle of light is greater than the critical angle, not less than the critical angle.

Educational objective:
Total internal reflection results in the light of an optical fiber traveling only within the core. Total internal reflection occurs when the incident angle of the light at the boundary between the core and the cladding is greater than the critical angle.

Time spent:0 sec
Subject:
Physics

QID:403518
System:
Light and Sound

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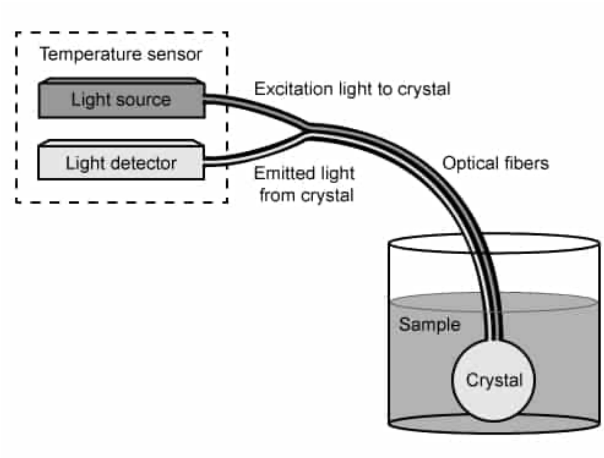

Figure 1 Schematic of a fiber optic temperature probe

Light is transmitted along continuous glass fibers that are approximately the width of a human hair. The fiber consists of a core surrounded by a cladding with a lower refractive index than the core. The difference in refractive indexes causes total internal reflection of the light at the interface between the core and the cladding, minimizing the loss of light intensity even over relatively long distances. One bundle of optical fibers transmits the excitation light to the crystal, and a separate bundle of optical fibers transmits the emitted light back to the device for analysis.

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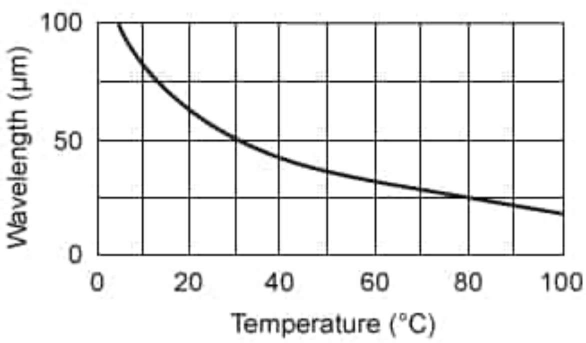


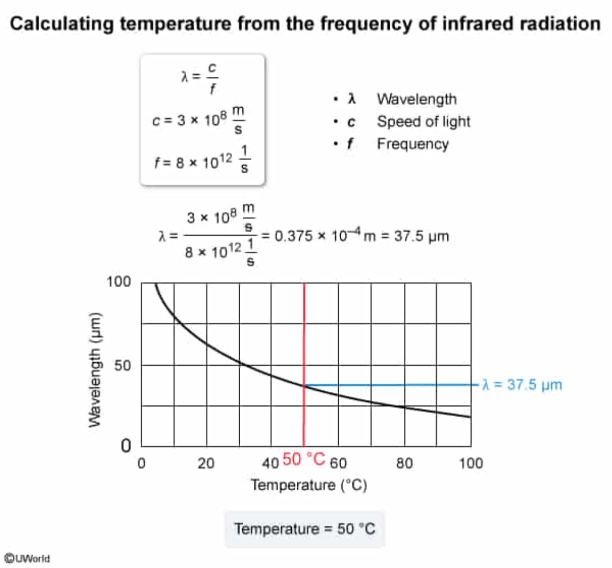
Figure 2 Plot of the infrared wavelength emitted at different temperatures

The frequency of infrared radiation detected from an object is 8×10^{12} Hz. What is the temperature of the object? (Note: For the speed of light, use 3×10^8 m/s.)

- A. 80 °C [14%]
- B. 50 °C [62%]
- C. 30 °C [18%]
- D. 10 °C [4%]

Correct
62% answered correctly

Explanation:



The wavelength λ of electromagnetic radiation equals the ratio of the speed of light c and the frequency f of the radiation:

$$\lambda = \frac{c}{f}$$

In this question, the f of the infrared radiation is 8×10^{12} Hz and the λ is calculated from the equation above:

$$\lambda = \frac{3 \times 10^8 \frac{\text{m}}{\text{s}}}{8 \times 10^{12} \frac{1}{\text{s}}}$$

$$\lambda = 0.375 \times 10^{-4} \text{ m}$$

$$\lambda = 37.5 \times 10^{-6} \text{ m} = 37.5 \ \mu\text{m}$$

The λ of infrared radiation at different temperatures is plotted in Figure 2 of the passage. From that data, when λ equals 37.5 μm, the temperature is 50 °C.

(Choice A) 80 °C is obtained by incorrectly dividing *f* by *c* and misinterpreting the result for λ as 27 μm.

(Choice C) 30 °C is obtained by incorrectly calculating λ as 50 μm.

(Choice D) 10 °C is obtained by incorrectly assuming λ is 80 μm.

Educational objective:
The wavelength of electromagnetic radiation equals the ratio of the speed of light and the frequency.

Time spent:0 sec
Subject:
Physics

QID:403519
System:
Light and Sound

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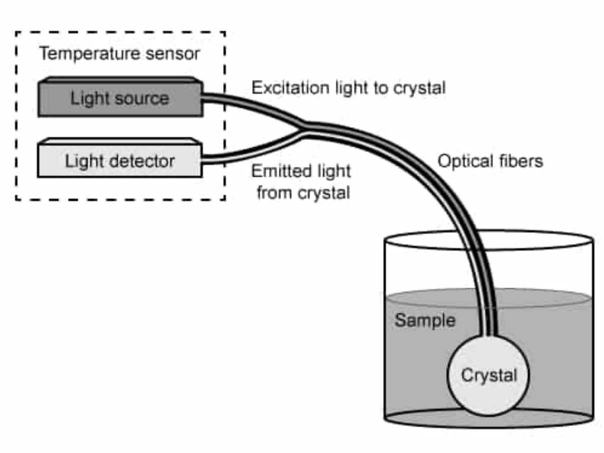

Figure 1 Schematic of a fiber optic temperature probe

Light is transmitted along continuous glass fibers that are approximately the width of a human hair. The fiber consists of a core surrounded by a cladding with a lower refractive index than the core. The difference in refractive indexes causes total internal reflection of the light at the interface between the core and the cladding, minimizing the loss of light intensity even over relatively long distances. One bundle of optical fibers transmits the excitation light to the crystal, and a separate bundle of optical fibers transmits the emitted light back to the device for analysis.

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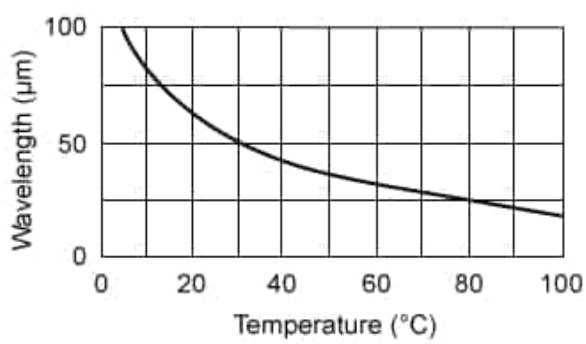


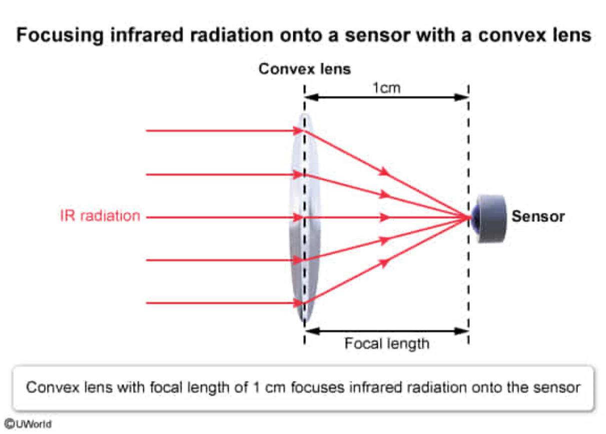
Figure 2 Plot of the infrared wavelength emitted at different temperatures

An infrared thermometer uses a lens to focus the incoming radiation onto a small sensor. What kind of lens should be used when the sensor is 1 cm behind the lens?

- A. Convex lens with a focal length of 1 cm [38%]
- B. Convex lens with a focal length of 2 cm [30%]
- C. Concave lens with a focal length of 1 cm [17%]
- D. Concave lens with a focal length of 2 cm [14%]

Correct
38% answered correctly

Explanation:



Convex lens with focal length of 1 cm focuses infrared radiation onto the sensor

A convex lens is shaped so that the center of the lens is thicker than the edges. As a result, parallel rays of light passing through a convex lens are refracted such that they bend toward each other; thus, a convex lens is also called a converging lens. The light rays cross each other and are focused at a distance equal to the **focal length** of the lens.

Conversely, a concave lens is thinner at the center than at the edges, causing parallel rays of light passing through the lens to be bent away from each other; thus, a concave lens is also called a diverging lens. The diverging light rays never cross each other and are not focused.

In this question, infrared radiation is focused on a sensor 1 cm behind the lens. Therefore, a convex lens is needed to focus the infrared rays after they pass through the lens, and a focal length of 1 cm is needed to focus the infrared rays onto the sensor.

(Choice B) A convex lens with a focal length of 2 cm would cause the infrared radiation to converge 2 cm behind the lens. The radiation would not be focused on the sensor because the sensor is only 1 cm behind the lens.

(Choices C and D) A concave lens is a diverging lens. The parallel rays of infrared radiation would spread out after passing through the lens instead of focusing onto the sensor.

Educational objective:
A convex lens causes light rays to converge at the focal length. A concave lens causes light rays to diverge.

Time spent:0 sec	**QID**:403520
Subject:	**System:**
Physics	Light and Sound

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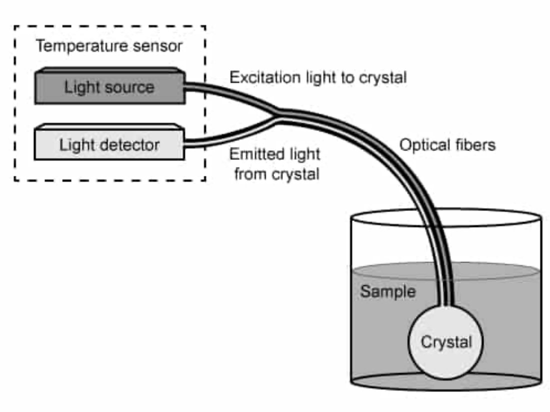

Figure 1 Schematic of a fiber optic temperature probe

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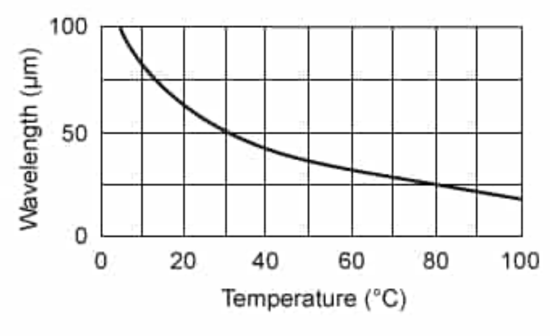


Figure 2 Plot of the infrared wavelength emitted at different temperatures

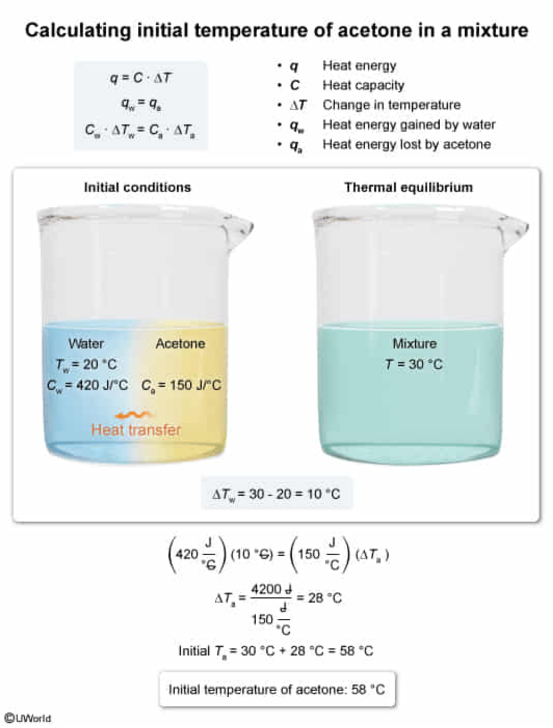
Water with an initial temperature of 20 °C is mixed with acetone in an insulated container.

The temperature inside the container is monitored with a fiber optic temperature probe, and the mixture reaches thermal equilibrium at 30 °C. The heat capacity of the water is 420 J/°C, and the heat capacity of the acetone is 150 J/°C. What was the initial temperature of the acetone?

- A. 2 °C [23%]
- B. 34 °C [18%]
- C. 40 °C [15%]
- D. 58 °C [42%]

Correct
42% answered correctly

Explanation:



The **heat energy** q of a material is equal to the product of the **heat capacity** C and the change in temperature ΔT:

$$q = C \cdot \Delta T$$

When two materials are mixed, one material loses heat q_1 and the other gains heat q_2 until they reach thermal equilibrium. If the mixture is thermally insulated, q from outside sources cannot be added or removed from the mixture, and q_1 will eventually equal q_2:

$$q_1 = q_2$$

Substituting the expression for q from the first equation yields:

$$C_1 \cdot \Delta T_1 = C_2 \cdot \Delta T_2$$

In this question, the temperature of the water increased from 20 °C to 30 °C. Thus, the change in temperature ΔT_w of the water is given by:

$$\Delta T_w = 30\ ℃ - 20\ ℃ = 10\ ℃$$

Substituting the values for the heat capacity C_w of water and the heat capacity C_a of acetone into the above equation gives:

$$420\frac{\text{J}}{^\circ\text{C}} \cdot 10\ ^\circ\text{C} = 150\frac{\text{J}}{^\circ\text{C}} \cdot \Delta T_a$$

Rearranging to solve for the change in temperature ΔT_a of the acetone yields:

$$\Delta T_a = \frac{420\frac{\cancel{\text{J}}}{\cancel{^\circ\text{C}}} \cdot 10\ ^\circ\text{C}}{150\frac{\cancel{\text{J}}}{\cancel{^\circ\text{C}}}} = 28\ ^\circ\text{C}$$

The water gained q after mixing with the acetone, so the acetone must have lost q. Therefore, the initial temperature T_a of the acetone is given by:

$$T_a = 30\ ^\circ\text{C} + \Delta T_a = 30\ ^\circ\text{C} + 28\ ^\circ\text{C} = 58\ ^\circ\text{C}$$

(Choice A) 2 °C is obtained by incorrectly subtracting ΔT_a from the temperature at equilibrium.

(Choice B) 34 °C is obtained by incorrectly using 150 J/°C for the heat capacity of water and 420 J/°C for the heat capacity of acetone.

(Choice C) 40 °C is obtained by incorrectly assuming that ΔT_a equals 10 °C.

Educational objective:
When two materials are combined and thermally insulated, the mixture will reach thermal equilibrium. The heat lost by one material is equal to the heat gained by the other material. The final temperature at equilibrium depends on the heat capacities and initial temperatures of the materials.

Time spent:0 sec
Subject:
Physics

QID:403521
System:
Light and Sound

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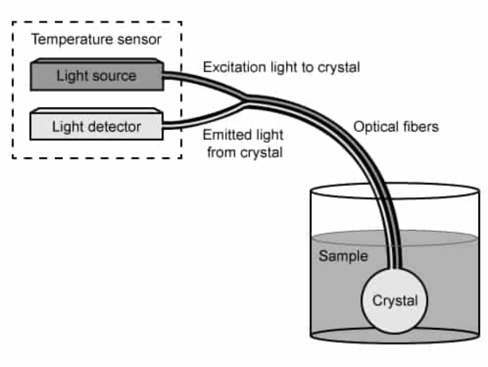

Figure 1 Schematic of a fiber optic temperature probe

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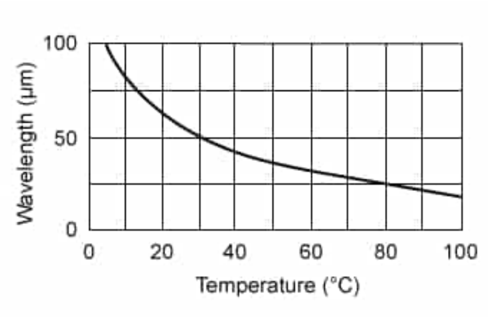


Figure 2 Plot of the infrared wavelength emitted at different temperatures

What is the approximate temperature of an ideal gas that has an average molecular kinetic energy of 7 × 10^{-21} J? (Note: For Boltzmann's constant, use k_B = 1.4 × 10^{-23} J/K.)

A. −20 °C [5%]

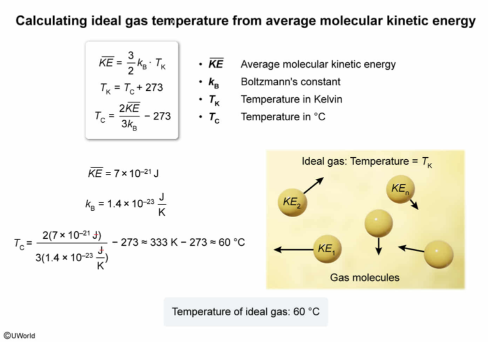
B. 60 °C [28%]

C. 230 °C [44%]

D. 330 °C [21%]

Correct
28% answered correctly

Explanation:



The kinetic molecular theory of gasses states that the average molecular **kinetic energy** $\overline{\text{KE}}$ of an ideal gas is proportional to the product of Boltzmann's constant k_B and the **temperature** T_K in Kelvin according to the equation:

$$\overline{\text{KE}} = \frac{3}{2} k_B \cdot T_K$$

This equation can be rearranged to solve for T_K:

$$T_K = \frac{2\overline{\text{KE}}}{3k_B}$$

The temperature T_C in Celsius is calculated from T_K using the equation:

$$T_C = T_K - 273$$

Substituting the equation for T_K into this relation yields:

$$T_C = \frac{2\overline{\text{KE}}}{3k_B} - 273$$

In this question, the value of $\overline{\text{KE}}$ for the gas and k_B are given. Substituting these values into the equation above gives:

$$T_C = \frac{2\left(7 \times 10^{-21}\ \cancel{\text{J}}\right)}{3\left(1.4 \times 10^{-23} \frac{\cancel{\text{J}}}{\text{K}}\right)} - 273$$

$$T_C = \frac{14 \times 10^{-21}}{4.2 \times 10^{-23}} \text{K} - 273$$

$$T_C \approx 3.33 \times 10^{-21+23}\ \text{K} - 273$$

$$T_C \approx 333\ \text{K} - 273 \approx 60\ \ °\text{C}$$

(Choice A) −20 °C is obtained by incorrectly calculating T_K as half the ratio of $\overline{\text{KE}}$ and k_B.

(Choice C) 230 °C is obtained by incorrectly calculating T_K as the simple ratio of $\overline{\text{KE}}$ and k_B.

(Choice D) 330 °C is obtained by incorrectly leaving the temperature in Kelvin instead of converting it to Celsius.

Educational objective:
The kinetic molecular theory of gasses is used to calculate the temperature of an ideal gas from the average kinetic energy of the gas molecules.

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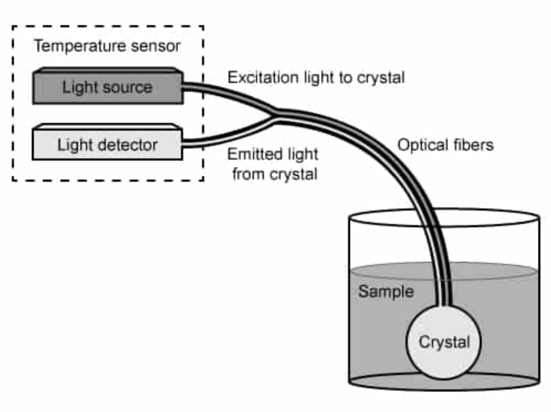

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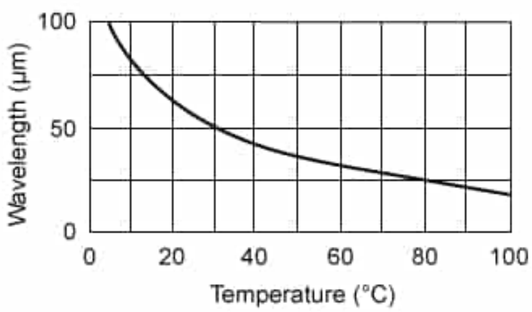


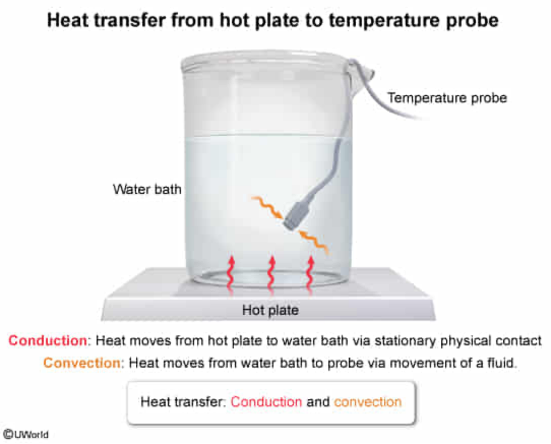
Figure 2 Plot of the infrared wavelength emitted at different temperatures

Which of the following best describes the order of the heat transfer mechanisms that take place when a fiber optic probe is placed in a water bath on a hot plate?

- A. Convection then conduction [21%]
- B. Radiation then convection [10%]
- ✔ C. Conduction then convection [47%]
- D. Conduction then radiation [20%]

Correct
47% answered correctly

Explanation:



Heat energy exchanges from warmer objects to cooler objects via three different heat transfer mechanisms: **conduction**, **convection**, and **radiation**. Conduction transfers heat between two objects by direct physical contact. Convection transfers heat through the flow of fluids. Radiation transfers heat through electromagnetic waves.

In this question, the hot plate generates heat that is transferred to the water bath and then to the fiber optic probe. The hot plate is in direct physical contact with the water bath, and the water transfers the heat to the fiber optic probe. Therefore, the heat transfer mechanisms are conduction (hot plate to water bath) then convection (water bath to probe).

(Choice A) Heat is not transferred from the hot plate to the water bath by convection, or from the water bath to the fiber optic probe by conduction.

(Choice B) Heat is not transferred from the hot plate to the water bath via radiation. The hot plate is in direct physical contact with the water bath, so heat is transferred by conduction.

(Choice D) Heat is transferred by conduction from the hot plate to the water bath. However, radiation does not transfer heat from the water bath to the fiber optic probe. The heat is transferred via convection because the movement of water transfers heat to the fiber optic probe.

Educational objective:
Heat transfer occurs by conduction, convection, or radiation. Conduction requires direct contact; convection requires fluid flow; and radiation requires emission of electromagnetic waves.

Time spent:0 sec **QID:**403523
Subject: **System:**